

Lesson 1: Create PT Deep Dive

45 minutes

Overview

This lesson contains a series of activities you can use to help students familiarize themselves with Create Performance Task and how it is scored. This lesson is also designed to introduce the nuances of the Create PT, and begin to provide some answers to the questions that will inevitably arise. The Create PT is in many ways straightforward: you complete a self-directed programming project and respond to prompts about your program and process. As you dig into the details of the task, however, you quickly come across some of the nuances of individual components of the task and how they're scored.

Agenda

Warm Up (10 minutes)

Introduce the Create PT

Review Create PT Submission Requirements

Activity (30 minutes)

The Create PT Survival Guide

Task Overview

What is Required of My Program?

Function Requirement Activity 1: Does It Count?

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Describe the scoring guidelines for the Create PT
- Evaluate sample Create PT components by applying the scoring guidelines

Preparation

- Print or prepare to distribute digital copies of the Create PT - Survival Guide
- Review pages 1-5 of the Create PT - Survival Guide
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our Virtual Lesson

Modifications

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 9 - Lesson 1 - Deep Dive** - Slides [▼ Make a Copy](#)
- **Create PT Prep Unit Slides** - Slides [▼ Make a Copy](#)

For the students

- **AP Computer Science Principles: Student Handouts** - Resource ([Download](#))

Teaching Guide

Warm Up (10 minutes)

Introduce the Create PT

Remarks

Today we're going to start looking more deeply at the Create PT, focusing specifically on understanding:

- The different components of the Create PT
- How the task will be scored

Don't worry, you already have much of the knowledge and skills you need to do well on this task. The hardest part might be just understanding what is required of you.

First, we'll quickly read the task description and look at some examples and how they were scored.

Teaching Tip

The Create PT is in many ways straightforward: you complete a self-directed programming project and respond to prompts about your program and process. As you dig into the details of the task, however, you quickly come across some of the nuances of individual components of the task and how they're scored. This lesson is designed to introduce what these nuances are, and begin to provide some answers to the questions that will inevitably arise. Keep in mind that the next lesson provides a more structured set of responses to those questions, and so today students are just diving in to what the task looks like.

Review Create PT Submission Requirements

 **Distribute:** Students should each get printed or digital copies of the Task Instructions.

 **Discuss:** *Read and then discuss with a partner the Task Instructions.*

Discussion Goal: **Aim to keep this discussion relatively short.** Assure students you're intending to log their questions and they will be addressed through the lesson.

Students should leave this discussion knowing they will submit:

- Video of their code running
- PDF of program code
- Personalized Program Reference including code snippets from their programs

Students are not, however, expected to fully understand the nuances of the task or scoring.

After reading discuss with a partner:

- What will you *actually* be turning in to the College Board?
- What are you hoping will become more clear before starting the project?

Discuss: *Give students time to read the pages, in pairs or individually, and then discuss both questions with one another. The first one is more important for now.*

Remarks

 Hopefully in your reading you concluded that for the Create PT you'll need to submit:

- Video showing your programs main functionality, including input and output
- A PDF of your program code
- Personalize Program Reference

I'm sure that right now you have a lot of questions about what this task will look like and how it will be scored. Before we answer them, let's look at some examples first.

Activity (30 minutes)

The Create PT Survival Guide

Remarks

Today we will use the Create PT Survival Guide to dig in a little deeper with the Create PT. The beginning of the packet has a few quick activities that help get us in the right mindset for thinking about and doing the task so you are ready to hit the ground running.

The guide will be useful throughout the entire process of completing the actual Create Task as well.

 **Distribute:** Create PT Survival Guide.

Task Overview (5 minutes)

 Have students quickly review the information on page 1 which summarizes components of the task and the role of the guide. Answer any high-level questions that come up.

What is Required of My Program? (10 minutes)

 Have students read the summary of the program requirements on page 2. The goal of this section should be to highlight the four takeaways in the Survival Guide. These should be familiar ideas to students from reviewing sample submissions in the previous lesson.

Remarks

Based on these four takeaways, it's clear that you have a lot of freedom in deciding how to design your program. Before moving on, we're going to complete two short activities to make sure we really understand the third requirement about our functions, since that one is the most complicated.

Function Requirement Activity 1: Does It Count? (15 minutes)

 Ask students to complete the activity on page 3, using the scoring guidelines on page 4. For each of the provided algorithms they need to decide whether the function would earn each row. Give students roughly 10 minutes to score each of the functions and then take 5 minutes to discuss their work and the main takeaways from this activity.

Discuss: Have students share and compare their responses with a classmate. Afterwards lead a discussion on the patterns they see.

Discussion Goal: The main takeaways from this activity should be:

- Some "complicated" looking functions may still not earn both rows if they are missing a component.
- Two functions may look very similar but because of small differences, one will earn full credit while the other will not.

Use the Scoring Guide Create PT Survival Guide - KEY for commentary on individual algorithms.

Wrap Up (5 minutes)

 **Discuss:** *What aspects of writing functions for the Create PT do you feel clicked today? What do you still have questions about?*

Have students share with one another before sharing with the whole class.

Dicsussion Goal: Use this opportunity to address any lingering questions or misconceptions in the room. You can also use this as a source of discussion topics to kick off the following lesson. As you lead the discussion, call out the many resources students have access to help when they're getting stuck.

Remarks

Next time we meet we'll look at strategies for narrowing down your topic and start to make a plan for completing the task.



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